A Study on Volatility Forecast of Wheat Futures Using a Mix of Econometric and Deep Learning Methods

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Abstract

This study evaluates the predictive efficacy of integrating various Generalized Autoregressive Conditional Heteroscedasticity (GARCH) family models—specifically standard GARCH, exponential GARCH (EGARCH), and exponentially weighted moving average (EWMA)—with advanced deep learning architectures for forecasting wheat futures volatility on the Chicago Mercantile Exchange. Furthermore, the performance of these hybrid frameworks is benchmarked against standalone econometric models and pure deep learning approaches, including Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks.

Contents

1 Introduction **4**

2 Related Work and Modeling Approaches **5**
2.1 Classical Volatility Models . . . 5
2.2 Econometric Models . . . 5
2.3 Machine Learning Approaches . . . 6
2.4 Hybrid Modeling Approaches . . . 6
2.5 Summary Table of Prior Studies . . . 7
2.6 Real-World Context . . . 7

3 Dataset Description **8**
3.1 Data Description . . . 8
3.2 Preprocessing and Cleaning . . . 10
3.3 Feature Engineering . . . 10
3.4 Train, Validation, and Test Splits . . . 11

4 Methodology and Model Specification **11**
4.1 Overview of Approaches . . . 11
4.2 Return Construction and Preprocessing . . . 12
4.3 GARCH(1,1) Model . . . 12
4.4 EGARCH Model . . . 13
4.5 EWMA Model . . . 13
4.6 Recurrent Neural Network . . . 13
4.6.1 Long Short-Term Memory . . . 14
4.7 Proposed Hybrid Models . . . 15

5 Methodology and Evaluation Metrics **15**
5.1 Realized Volatility . . . 15
5.2 Loss Functions . . . 15
5.3 Test for Comparing Forecast Accuracy . . . 16
5.4 Experimental Design . . . 17

6 Model Training and Hyperparameters **17**
6.1 GARCH(1,1) Estimation . . . 17
6.2 RNN Training Setup . . . 18
6.3 Hyperparameter Tuning . . . 18

7 Results **19**
7.1 Plots . . . 20

8 Model Comparison and Robustness Checks **20**
8.1 Model Comparison . . . 20
8.2 Robustness Checks . . . 20

9 Risks & Limitations **22**

10 Challenges Faced **22**

11 Conclusion **23**

A Additional Figures **24**

1 Introduction

Commodity markets play a vital role in the global economy by facilitating price discovery and risk transfer for producers, consumers, and investors. Agricultural commodities such as wheat, soybeans, and corn are subject to significant price fluctuations due to factors including weather conditions, geopolitical events, supply–demand imbalances, and macroeconomic uncertainty. These fluctuations introduce substantial risk for market participants, making effective risk management strategies essential. In practice, commodities are predominantly traded through futures contracts rather than direct spot market transactions. Although spot prices may not always be actively traded on exchanges, they represent the underlying economic value of the commodity and serve as a fundamental reference for futures pricing. As futures contracts approach their expiration date, their prices converge toward the prevailing spot price, linking futures market dynamics closely to underlying price behavior. Consequently, understanding and forecasting price volatility remains crucial for hedging, pricing derivatives, and managing exposure in commodity futures markets. Volatility forecasting has long been a central topic in financial econometrics. Classical models such as the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) family have been widely used to capture time-varying volatility and volatility clustering observed in financial and commodity return series. Among these, the GARCH(1,1) model remains a standard benchmark due to its simplicity, interpretability, and strong empirical performance. In recent years, advances in machine learning and deep learning have introduced alternative approaches to time-series modeling. Recurrent Neural Networks (RNNs), in particular, are well suited for sequential data as they can learn temporal dependencies and nonlinear patterns that may not be adequately captured by traditional econometric models. These characteristics make RNNs a promising tool for modeling complex dynamics in commodity price movements and volatility.
In summary, this study conducts a rigorous comparative analysis between classical econometric frameworks and deep learning architectures for forecasting wheat futures volatility. Specifically, we evaluate the predictive power of GARCH-family models—including standard GARCH, EGARCH, and EWMA—alongside Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) units. By further developing hybrid models that integrate GARCH-type specifications with RNN architectures, this research seeks to determine whether combining statistical rigor with machine learning flexibility offers superior accuracy in capturing the complex dynamics of agricultural commodity markets.

We conduct the study in the following steps:

- Model wheat futures volatility using the GARCH family of models to create a baseline.
- Build standalone deep learning models using RNNs and LSTMs.
- Develop hybrid models by adding a single GARCH feature to the RNN and LSTM architectures.
- Expand the hybrid models by including multiple GARCH features to see if it improves accuracy.
- Compare the results across all models to determine which approach provides the best fit for the data.

The remainder of this report is organized as follows: Section 2 summarizes relevant literature. Section 3 presents the dataset and preprocessing steps. Section 4 describes the GARCH(1,1) and RNN model specifications. Section 6 details the training procedure and hyperparameters. Section 7 reports empirical results post which we show plots of the performace Section 7.1.

2 Related Work and Modeling Approaches

2.1 Classical Volatility Models

Volatility plays a crucial role in commodity markets, influencing hedging strategies, risk management, and derivative pricing. For agricultural commodities like wheat, price fluctuations can be affected by factors such as weather conditions, global supply-demand imbalances, and policy interventions. Accurate volatility estimation is essential for producers, traders, and policymakers to make informed decisions and mitigate risk (Haase & Henn, 2024). A standard GARCH(1,1) model for a return series $\{r_t\}$ can be written as:

$$r_t = \mu + \epsilon_t, \tag{1}$$

$$\epsilon_t = \sigma_t z_t, \quad z_t \sim \text{i.i.d. } (0,1), \tag{2}$$

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2. \tag{3}$$

The parameters (ω, α, β) capture the long-run variance level and the persistence of volatility shocks.

2.2 Econometric Models

Traditional econometric models have been widely employed to analyze and forecast volatility in financial and commodity markets. These models are designed to capture time-varying variance and the phenomenon of volatility clustering, whereby periods of high volatility tend to persist over time. One of the earliest and most influential approaches is the Autoregressive Conditional Heteroskedasticity (ARCH) model introduced by Engle (1982), which represents conditional variance as a function of past squared innovations. While ARCH provided a foundational framework for modeling heteroskedasticity, it often required a large number of parameters to adequately capture volatility persistence.

To overcome this limitation, Bollerslev (1986) proposed the Generalized ARCH (GARCH) model, which extends the ARCH framework by allowing conditional variance to depend on both lagged squared shocks and past conditional variances. This extension improves model parsimony and forecasting performance, making GARCH a standard benchmark in volatility modeling. Building upon the GARCH framework, the Exponential GARCH (EGARCH) model further enhances flexibility by modeling the logarithm of conditional variance. This formulation removes the need for non-negativity constraints on model parameters and enables the explicit modeling of asymmetric effects, where positive and negative shocks have differing impacts on volatility. Such leverage effects are particularly relevant in commodity markets, where adverse news or supply-side shocks may trigger stronger volatility responses.

Another commonly used approach is the Exponentially Weighted Moving Average (EWMA) model, which estimates volatility by assigning exponentially decreasing weights to past squared returns. EWMA provides a parsimonious and computationally efficient method for volatility estimation and is widely applied in risk management settings. Although these econometric models are effective in capturing volatility persistence and conditional heteroskedasticity, they rely on assumptions of stationarity, linear dependence structures, and predefined functional forms. These assumptions can restrict their ability to adequately model the complex, nonlinear, and regime-dependent dynamics present in agricultural commodity markets. In the case of wheat prices, volatility is influenced by interacting factors such as climatic conditions, geopolitical events, supply chain disruptions, and policy interventions, which introduce nonlinearities that traditional econometric models may struggle to capture fully (Haase & Henn, 2024).

The ARCH, GARCH, EGARCH, and EWMA models are selected as baseline approaches in this study due to their well-established theoretical foundations and widespread use in volatility modeling across financial and commodity markets. These models provide interpretable benchmarks for capturing key stylized facts such as volatility clustering, persistence, and, in the case of EGARCH, asymmetric shock responses. Using these econometric frameworks as baselines allows for a systematic evaluation of model performance and ensures that any observed improvements from machine learning–based approaches are meaningful rather than artifacts of model complexity. Moreover, EWMA offers a parsimonious risk-oriented benchmark, while GARCH-family models serve as industry-standard references, making them particularly suitable for comparative analysis. Establishing these traditional models as baselines thus provides both methodological rigor and practical relevance before introducing more flexible, data-driven machine learning models.

2.3 Machine Learning Approaches

Machine learning approaches have increasingly been applied to commodity price and volatility forecasting due to their ability to model complex, nonlinear relationships without imposing restrictive parametric assumptions. Unlike traditional econometric models, machine learning methods are data-driven and can adapt to structural changes and regime shifts commonly observed in agricultural markets.

Recurrent Neural Networks (RNNs), a class of Artificial Neural Networks, are specifically designed to model sequential and time-dependent data by incorporating feedback connections that allow information to persist across time steps. Unlike feedforward architectures, RNNs maintain an internal hidden state that captures temporal dependencies, enabling the network to learn patterns that evolve over time. This makes RNNs particularly effective for modeling financial time-series data, such as historical prices, returns, and volatility measures, where past observations influence future behavior. By explicitly accounting for temporal dynamics, RNNs provide a more suitable framework for capturing sequential relationships compared to traditional feedforward networks.

Recurrent Neural Networks (RNNs) address this limitation by explicitly incorporating temporal dependencies through recurrent connections. Among RNN variants, Long Short-Term Memory (LSTM) networks have emerged as a robust architecture for time-series forecasting. LSTMs are designed to capture both short-term dynamics and long-range dependencies while mitigating issues such as vanishing and exploding gradients. This makes them particularly suitable for volatility modeling, where past shocks and persistent effects play a critical role. Empirical evidence suggests that LSTM models often outperform traditional econometric benchmarks in environments characterized by nonlinear dynamics and structural variability.

Recent literature has also explored the use of alternative deep learning architectures, such as Convolutional Neural Networks combined with satellite imagery and weather indicators, to improve wheat price forecasting (Thaker, Chan, and Sonner, 2024). While such approaches highlight the importance of incorporating external information, this study focuses on RNN and LSTM models to evaluate the extent to which purely data-driven neural architectures can improve volatility forecasting performance relative to traditional econometric baselines.

2.4 Hybrid Modeling Approaches

To systematically evaluate the impact of hybridization between econometric and deep learning models, this study considers multiple model configurations ranging from single-model benchmarks to progressively enriched hybrid architectures. The single-model category includes traditional econometric models such as EWMA, GARCH, and EGARCH, as well as standalone machine learning

models, namely the Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) network. These models serve as baseline references for assessing the incremental benefits of hybrid modeling.

Building upon these baselines, RNN-based hybrid models are constructed by augmenting the RNN with volatility estimates generated from a single GARCH-type model. Specifically, EWMA-, GARCH-, and EGARCH-based conditional variance estimates are used as additional inputs to the RNN, resulting in W-RNN, G-RNN, and E-RNN configurations, respectively. This design allows the RNN to leverage econometric volatility structure while learning nonlinear relationships from data.

Similarly, LSTM-based hybrid models are developed to incorporate econometric information into a sequence-learning framework. In the first hybrid layer, LSTM models are combined with one GARCH-type volatility estimate, producing W-LSTM, G-LSTM, and E-LSTM architectures. These models assess whether the inclusion of a single econometric volatility signal improves temporal learning and forecasting accuracy.

To further investigate the effect of increasing econometric information, additional LSTM-based hybrids incorporate two and three GARCH-type volatility measures simultaneously. The two-model hybrids, including GW-LSTM, EW-LSTM, and GE-LSTM, integrate pairs of volatility estimates, while the three-model hybrid GEW-LSTM combines EWMA, GARCH, and EGARCH outputs within a single LSTM framework. These progressively enriched configurations enable an examination of whether multiple econometric signals provide complementary information or introduce redundancy Kim and Won (2018).

By evaluating this structured set of model combinations, the study aims to assess not only the effectiveness of hybrid modeling relative to standalone approaches, but also the optimal degree of hybridization for volatility forecasting in wheat markets. This hierarchical design provides a clear and interpretable framework for understanding how econometric structure and deep learning flexibility interact to improve predictive performance.

2.5 Summary Table of Prior Studies

Table 1: Summary of Selected Studies on Volatility Modeling in Financial and Commodity Markets

Study	**Year**	**Methodology**	**Commodity/Market**	**Key Findings**
Engle (1982)	1982	ARCH	Stock market	Introduced volatility modeling with conditional variance
Bollerslev (1986)	1986	GARCH	Stock market	Captured volatility clustering, improved over ARCH
Chan et al. (2024)	2024	CNN + Satellite data	Wheat futures	Improved price prediction accuracy using ML
Haase and Henn (2024)	2024	GARCH + ANN	Wheat futures	Hybrid model reduced forecast error

2.6 Real-World Context

Volatility forecasting plays a critical role in wheat markets, as wheat prices are subject to substantial fluctuations driven by seasonality, weather-related shocks, geopolitical developments, and global supply–demand imbalances. These factors introduce uncertainty and instability in price movements, making accurate volatility estimation essential for effective price risk management in agricultural commodity markets. Reliable volatility forecasts enable market participants to anticipate periods of heightened risk and make informed decisions under uncertain conditions. For

farmers and agricultural producers, volatility estimates are particularly valuable in supporting operational and strategic decision-making. Accurate forecasts assist in planning key activities such as sowing, harvesting, storage, and determining the optimal timing for market participation. Improved volatility forecasting can significantly reduce income uncertainty by mitigating the adverse effects of sudden and unpredictable price movements, thereby contributing to more stable revenue streams and improved financial resilience.

Volatility forecasting is equally important for traders and hedgers, as wheat is predominantly traded through futures contracts rather than spot markets. In such derivative-based trading environments, volatility serves as a fundamental input for risk assessment, margin requirements, and hedging strategies. Reliable volatility predictions improve hedge timing, facilitate more effective hedge ratio selection, and enhance exposure management. Moreover, accurate volatility estimates contribute to more efficient futures pricing and overall risk control within commodity trading systems.

Traditional econometric models, including GARCH-type models and EWMA, have been widely used to estimate wheat price volatility due to their ability to capture volatility clustering and persistence. While these models offer strong theoretical foundations and interpretability, they rely on restrictive assumptions such as linear dependence structures and stationarity. As a result, they often exhibit limited capacity to model the nonlinear dynamics and long-term dependencies that are common in commodity price movements influenced by complex external factors.

Neural network–based models offer a promising alternative by providing greater flexibility in capturing nonlinear relationships. In particular, Long Short-Term Memory (LSTM) networks have demonstrated superior performance over feedforward neural networks by effectively learning long-range temporal dependencies in time-series data. This capability is especially relevant for wheat markets, which are characterized by seasonal patterns and delayed responses to weather conditions, policy changes, and global market information.

These considerations motivate the adoption of a hybrid modeling approach that integrates econometric and neural network methodologies. Purely econometric models may fail to capture complex nonlinear structures, while purely neural models may overlook economically meaningful volatility features embedded in traditional frameworks. By incorporating multiple GARCH-type model outputs and EWMA estimates as inputs to an LSTM network, the proposed hybrid framework combines theoretical volatility structure with deep learning flexibility. This integrated approach aims to enhance forecasting accuracy and improve risk management outcomes in wheat commodity trading and hedging applications.

3 Dataset Description

3.1 Data Description

The dataset consists of daily wheat futures market data obtained from a rolling futures . To ensure the reliability of our volatility forecasts, we utilize a rolling series of mid-expiry wheat futures rather than the front-month (near-expiry) contracts. This choice helps to mitigate the Samuelson Effect proposed by Samuelson (1965) which often causes artificial volatility spikes as a contract nears its expiration and physical delivery period. By using mid-expiry data, we capture more stable liquidity and avoid the technical price distortions typically associated with contract rolls, providing a cleaner signal for our GARCH and RNN models.

The sample spans from January 5, 1996, to December 16, 2025, comprising a total of 7,813 observations. This long historical coverage enables the analysis of volatility dynamics across multiple market regimes, including periods of heightened price instability driven by weather shocks, global

supply–demand imbalances, and geopolitical events.
Additonally, the daily closing prices of gold and global wheat production (Metric tons per month) data were taken from bloomberg to use as additional predictor variables for our neural networks. Table 2 shows common descriptive statistics as mean, standard deviation, skewness, and kurtosis of the time series data used in this study and also shows the results of the augmented Dickey–Fuller (ADF) test, which is a stationary test, and the Jarque-Bera test, which is a normality test. An ADF test was used to confirm the stability of this time series. The null hypothesis (h0) of the ADF test is that the correspond- ing time series has a unit root. As shown in Table 1, the ADF test statistic is -2.65, but the 5% critical value is -2.86, so the null hypothesis cannot be rejected , and hence the time series has a unit root, which is unstable (Dickey and Fuller, 1979). We transform the nonstationary into a by using a return value that takes the difference of the logarithmic value of the original time series.

In Table 2, the ADF test statistics for the logarithmic returns show -87.36, which is less than the 5% critical value of -2.56; therefore, the null hypothesis H_0 can be rejected. As shown in Figure 2, the logarithmic return time series oscillates around a mean of zero, indicating that the series is stationary.

As Figure 1 and Table 2 demonstrate, the distribution of Wheat Futures log returns reveals high kurtosis. In other words, the distribution has a thicker tail than the normal distribution. Additionally, the Jarque-Bera test for the logarithmic returns yielded a high statistic of 1666.69 with a p-value <0.001 . This indicates a significant departure from normality, primarily driven by the observed excess kurtosis of 2.22. These results confirm the presence of 'fat tails' and non-normal characteristics in wheat futures. This provides a strong justification for employing GARCH and RNN models, which are better suited than classical linear models for capturing such complex, non-linear distributions

Given the non-stationary nature of price levels in commodity markets, the analysis proceeds by transforming prices into log returns. This transformation stabilizes the variance and removes trend components. Preliminary diagnostic analysis reveals volatility clustering and heavy-tailed behavior, which motivates the application of GARCH-type models as well as deep learning approaches, such as Long Short-Term Memory (LSTM) networks, for volatility forecasting.

The dataset contains no missing observations, ensuring robustness in model estimation and reliability in out-of-sample evaluation.

Table 2: Summary Statistics and Diagnostic Test Results

Series	**Mean**	**Std Dev**	**Skew**	**Kurtosis**	**Jarque–Bera**	**ADF**
wheat_returns	517.8818	185.0947	0.7228	0.3425	718.48	-2.6459
gold_returns	1100.6219	741.2034	1.0497	1.6526	2323.83	4.7892
production_growth	2074.3538	248.3637	0.0437	-0.7720	196.48	-2.7499
log_returns	0.0013	1.8189	0.2071	2.2246	1666.69	-87.3599

Let P_t denote the closing price at time t. We compute log-returns as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right). \tag{4}$$

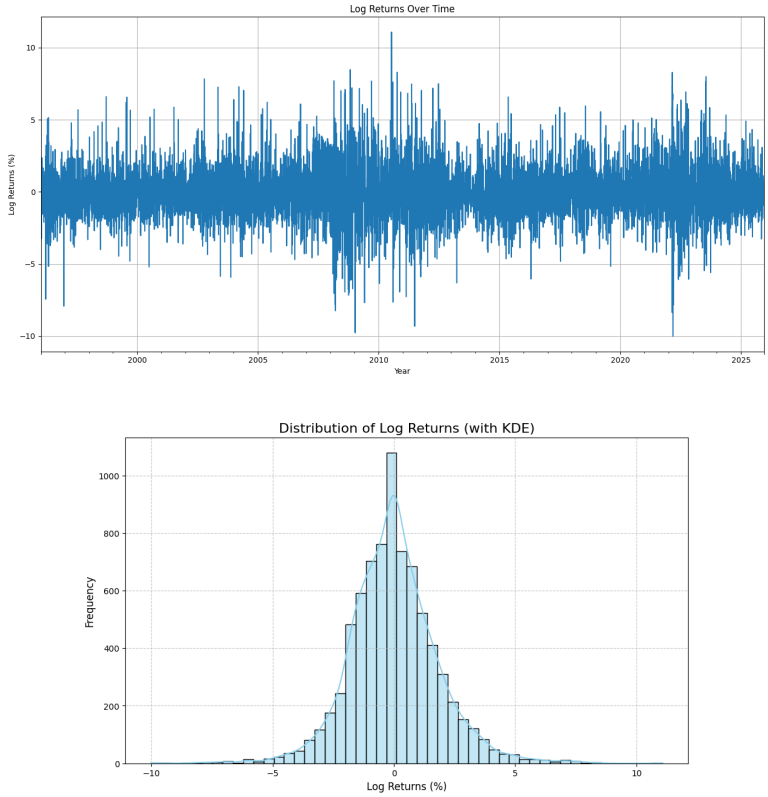


Figure 1: Log returns and related dynamics

3.2 Preprocessing and Cleaning

Prior to model estimation, several preprocessing steps were applied to ensure data quality and suitability for volatility modeling and deep learning approaches. First, the dataset was inspected to verify consistent daily trading timestamps. Non-trading days, such as weekends and exchange holidays, are naturally absent from futures market data and therefore required no additional filtering.

The dataset was examined for missing observations and any gaps were handled using forward-filling, and any Nan values were handled with forward filling .

Since out analysis focuses on volatilty predicction futures contracts rather than individual equities, adjustments for corporate actions such as stock splits or dividends were not applicable. Instead, a back-adjusted continuous futures series was used to eliminate artificial price jumps arising from contract rollover, thereby preserving economically meaningful return dynamics.

Finally, for models based on deep learning architectures such as Long Short-Term Memory (LSTM) networks, input features were standardized using z-score normalization. This normalization improves numerical stability and accelerates convergence during model training. Subsequent return construction and statistical diagnostics are described in Section 4.2

3.3 Feature Engineering

To enhance the predictive capability of the deep learning models, we perform feature engineering to capture diverse aspects of volatility and market dynamics. The following features are constructed for each time step t:

1. **Logarithmic Returns (ret_t):** The primary input variable representing the percentage change in wheat futures prices:

$$ret_t = 100 \times [\ln(P_t) - \ln(P_{t-1})] \tag{5}$$

2. **Volatility Proxies (abs_ret_t and ret_t^2):** Absolute and squared returns are used to capture the magnitude of price shocks:

$$abs_ret_t = |ret_t|, \quad ret_t^2 = (ret_t)^2 \tag{6}$$

3. **Log-Transformed Squared Returns (log_ret2_t):** To compress the scale of variance signals and improve neural network convergence, we apply a log-transformation:

$$log_ret2_t = \ln(ret_t^2 + \epsilon) \tag{7}$$

   where $\epsilon = 10^{-12}$ is a small constant used for numerical stability to avoid $\ln(0)$.

4. **External Market Drivers ($gold_ret_t$ and $prod_growth_t$):** We incorporate exogenous variables to capture broader economic trends. We take log returns of gold and production growth as they tend to be stable and add robustness in preditcion.

$$gold_ret_t = 100 \times [\ln(G_t) - \ln(G_{t-1})] \tag{8}$$

$$prod_growth_t = \ln(W_t) - \ln(W_{t-1}) \tag{9}$$

   where G_t represents the price of gold and W_t represents global wheat production.

This combined feature set allows the RNN and LSTM architectures to learn from both internal price history and broader macroeconomic indicators.

3.4 Train, Validation, and Test Splits

After data preprocessing, 7,802 valid observations were retained for the empirical analysis. The time series was partitioned into non-overlapping sequential data points to preserve the temporal ordering and prevent information leakage. Of these observations, 6,241 data points were assigned to the training sample, while the remaining 1,561 data points were reserved for out-of-sample evaluation.

The partitioning scheme is defined as follows:

- **Training set**: first 6,241 data points, used to estimate GARCH model parameters and to train the recurrent neural network (RNN) models.
- **Test set**: final 1,561 data points, held out and used exclusively for final model comparison and performance evaluation.

4 Methodology and Model Specification

4.1 Overview of Approaches

This study investigates volatility forecasting in the wheat commodity market using both econometric and deep learning approaches. Two primary model families are considered:

- Parametric volatility models, including GARCH(1,1), EGARCH, and EWMA, estimated using maximum likelihood methods.
- Recurrent Neural Network (RNN)-based models, specifically Long Short-Term Memory (LSTM) networks, trained using gradient-based optimization.

The primary target of interest is the one-step-ahead conditional volatility of wheat prices, proxied by squared returns or model-implied conditional variance, depending on the evaluation setting.

4.2 Return Construction and Preprocessing

To ensure the data were suitable for time-series modeling, daily log returns were constructed from wheat futures prices. Let P_t denote the settlement price at time t. The continuously compounded return is defined as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right).$$

Observations with missing values arising from differencing or rolling-window calculations were removed to maintain consistency in the time series. The resulting return series was examined to assess key statistical properties relevant for volatility modeling.

Stationarity of the log-return series was evaluated using visual inspection and standard statistical diagnostics. In addition, autocorrelation functions (ACF) of both squared and absolute returns were analyzed to identify volatility clustering, a stylized feature commonly observed in financial time-series data. The presence of volatility clustering provides empirical motivation for the use of conditional volatility models in subsequent analysis.

4.3 GARCH(1,1) Model

In classical regression analysis, it is commonly assumed that the variance of the error term is constant. Under the assumption that the probability distributions of the residuals are identical across observations, estimators remain unbiased; however, they are no longer efficient when conditional variance is time-varying. Consequently, statistical tests and confidence intervals derived under constant variance assumptions become unreliable.

To address this issue, Engle (1982) proposed the Autoregressive Conditional Heteroskedasticity (ARCH) model, which explicitly models time-varying volatility. The return process is defined as

$$y_t = \mu_t + \sigma_t \eta_t, \quad \eta_t \sim \mathcal{N}(0,1), \tag{10}$$

where μ_t represents the conditional mean and σ_t denotes the conditional standard deviation. The error term is expressed as

$$\varepsilon_t = \sigma_t \eta_t, \quad \varepsilon_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, \sigma_t^2), \tag{11}$$

where $\mathcal{F}_{t-1}$ denotes the information set available at time $t-1$.

In the ARCH(q) model, the conditional variance depends on past squared residuals:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2. \tag{12}$$

To ensure positivity of the conditional variance, all parameters are required to be nonnegative. A limitation of the ARCH model is the difficulty in selecting an appropriate lag order q, as higher orders require estimating a large number of parameters.

Bollerslev (1986) generalized the ARCH model by introducing the Generalized ARCH (GARCH) model, which incorporates lagged conditional variances:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2. \tag{13}$$

This formulation allows volatility to depend on both past shocks and past volatility levels.

The GARCH(1,1) model, which is widely used due to its parsimony and effectiveness, is given by

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \tag{14}$$

By rearranging the equation, the conditional variance can be expressed as an infinite weighted sum of past squared residuals:

$$\sigma_t^2 = \frac{\alpha_0}{1-\beta_1} + \alpha_1 \sum_{j=1}^{\infty} \beta_1^{j-1} \varepsilon_{t-j}^2. \tag{15}$$

This representation assigns geometrically decreasing weights to past shocks, consistent with the volatility clustering observed in financial and commodity time series.

4.4 EGARCH Model

The Exponential GARCH (EGARCH) model proposed by Nelson (1991) removes the non-negativity constraints imposed on GARCH parameters and explicitly captures asymmetric volatility effects. The return equation is defined as

$$r_t = X_t M + \varepsilon_t, \tag{16}$$

while the conditional variance follows

$$\ln \sigma_t^2 = \alpha_0 + \beta \ln \sigma_{t-1}^2 + \omega \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \gamma \left| \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \right|. \tag{17}$$

By modeling the logarithm of the variance, the EGARCH specification ensures positivity of σ_t^2 regardless of parameter signs. The parameter ω captures leverage effects, allowing negative and positive shocks of equal magnitude to have asymmetric impacts on volatility.

4.5 EWMA Model

The Exponentially Weighted Moving Average (EWMA) model, popularized by the RiskMetrics framework J.P. Morgan RiskMetrics Group (1996), defines conditional variance as:

$$\sigma_t^2 = \rho \sigma_{t-1}^2 + (1-\rho)\varepsilon_{t-1}^2, \quad 0 < \rho < 1. \tag{18}$$

This formulation places greater emphasis on recent observations, making the model responsive to short-term volatility changes. However, excessively large values of ρ may cause the model to ignore long-term volatility dynamics.

4.6 Recurrent Neural Network

Recurrent Neural Networks (RNNs) are a class of neural network models specifically designed to capture sequential dependencies in time-series data. Unlike feedforward networks, RNNs introduce

feedback connections that allow information from previous time steps to influence current predictions. This makes RNNs particularly suitable for modeling financial time series, where volatility exhibits temporal dependence.

At each time step t, the hidden state of an RNN is updated as:

$$h_t = f(W_x x_t + W_h h_{t-1} + b), \tag{19}$$

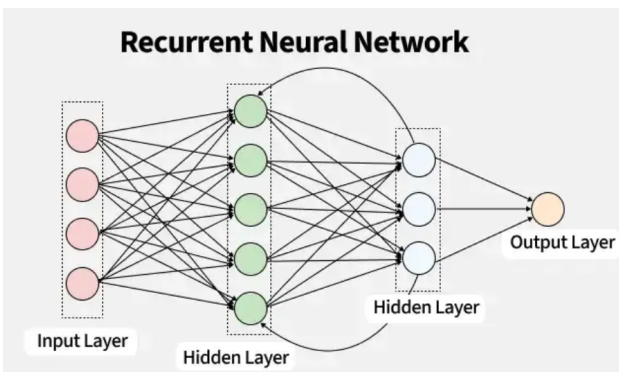
where x_t denotes the input at time t, h_{t-1} is the hidden state from the previous time step, W_x and W_h are weight matrices, b is a bias vector, and $f(\cdot)$ represents a nonlinear activation function.

The predicted volatility is obtained from the hidden state as:

$$\hat{v}_t = g(W_o h_t + b_o), \tag{20}$$

where $g(\cdot)$ is an output activation function chosen to ensure non-negative volatility forecasts.

RNNs are trained using supervised learning, with parameters optimized through backpropagation through time (BPTT) and gradient-based optimization methods Goodfellow et al. (2016). Despite their flexibility, standard RNNs may suffer from vanishing and exploding gradient problems when modeling long sequences, motivating the use of more advanced architectures such as Long Short-Term Memory (LSTM) networks.



4.6.1 Long Short-Term Memory

Recurrent Neural Networks (RNNs) are designed to learn sequential dependencies but suffer from vanishing and exploding gradient problems. RNNs by introducing memory cells and gating mechanisms Long Short-Term Memory (LSTM) networks overcome these limitations by introducing memory cells and gating mechanismsHochreiter and Schmidhuber (1997). The LSTM dynamics are defined as

$$g_t = \sigma(U_g x_t + W_g h_{t-1} + b_g), \tag{21}$$
$$i_t = \sigma(U_i x_t + W_i h_{t-1} + b_i), \tag{22}$$
$$\tilde{c}_t = \tanh(U_c x_t + W_c h_{t-1} + b_c), \tag{23}$$
$$c_t = g_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{24}$$
$$o_t = \sigma(U_o x_t + W_o h_{t-1} + b_o), \tag{25}$$
$$h_t = o_t \odot \tanh(c_t). \tag{26}$$

Here, $\sigma(\cdot)$ denotes the sigmoid function and $\odot$ represents element-wise multiplication. LSTMs effectively capture long-range temporal dependencies and are well suited for volatility forecasting in time-series data.

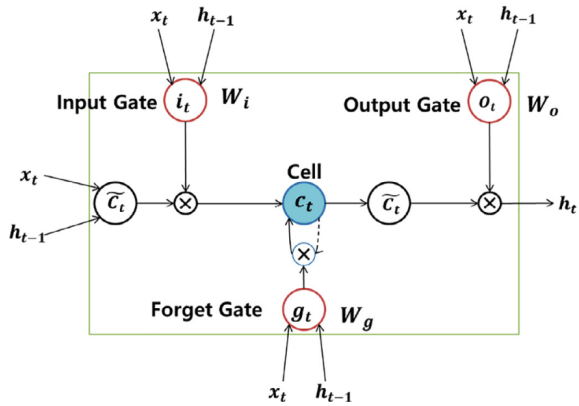


Figure 2: An LSTM memory block.

4.7 Proposed Hybrid Models

The proposed hybrid framework integrates GARCH-type models with neural networks to leverage both econometric structure and deep learning flexibility. Parameters estimated from GARCH, EGARCH, and EWMA models are used as inputs to RNN and LSTM architectures. By combining multiple volatility signals—such as shock magnitude, persistence, leverage effects, and short-term dynamics—the hybrid models aim to improve volatility forecasting performance relative to standalone econometric or neural network models.

5 Methodology and Evaluation Metrics

5.1 Realized Volatility

Using daily settlement prices of wheat futures, daily log returns are computed as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right).$$

Following Andersen et al. (2003), realized volatility is constructed using the sum of squared daily log returns over a 22-day horizon. Volatility is defined as the 22-day ahead realized volatility, constructed as the square root of the sum of squared daily log returns from $t+1$ to $t+22$:

$$\mathrm{Vol}_t^{(22)} = \sqrt{\sum_{i=1}^{22} r_{t+i}^2}.$$

This forward-looking measure serves as an ex post observable proxy for latent volatility and is used as the target variable in the supervised learning framework. Model performance is evaluated by comparing forecasted volatility with realized volatility, allowing us to assess the ability of the models to predict future risk rather than merely describe past variability.

5.2 Loss Functions

Model performance is evaluated using MAE, MSE, and the QLIKE loss function Patton (2011).To evaluate the forecasting performance of the volatility models, multiple loss functions are employed,

as different measures emphasize different aspects of forecast accuracy. Using a range of loss functions allows for a more robust comparison of competing models.

The mean absolute error (MAE) and mean squared error (MSE) measure the absolute and squared deviations between forecasted and realized volatility, respectively. While MAE is less sensitive to large forecast errors, MSE penalizes large deviations more heavily.

To account for the scale-dependent and heteroscedastic nature of volatility, we also consider heteroscedasticity-adjusted loss functions. The heteroscedasticity-adjusted MAE (HMAE) and heteroscedasticity-adjusted MSE (HMSE) evaluate relative forecast errors by normalizing forecast deviations with realized volatility.

The loss functions are defined as follows:

$$\text{MAE} = \frac{1}{T}\sum_{t=1}^{T} |\hat{v}_t - RV_t|\,, \tag{27}$$

$$\text{MSE} = \frac{1}{T}\sum_{t=1}^{T} (\hat{v}_t - RV_t)^2\,, \tag{28}$$

$$\text{HMAE} = \frac{1}{T}\sum_{t=1}^{T} \left|1 - \frac{\hat{v}_t}{RV_t}\right|, \tag{29}$$

$$\text{HMSE} = \frac{1}{T}\sum_{t=1}^{T} \left(1 - \frac{\hat{v}_t}{RV_t}\right)^2, \tag{30}$$

Here, $\hat{v}_t$ denotes the forecasted volatility at time t, RV_t represents the realized volatility, and T is the total number of out-of-sample forecasts.

5.3 Test for Comparing Forecast Accuracy

We performed the Diebold–Mariano (DM) test and Wilcoxon signed-rank (WS) test to verify the equivalence of forecast accuracy between two forecasting models Diebold and Mariano (1995) .

For the DM test, forecast errors are denoted as $u_{i,t} = \hat{y}_{i,t} - y_t$, for $i = 1, 2$, where y_t is the actual time series and $\hat{y}_{i,t}$ is the forecast. The null hypothesis of equal accuracy for the two models is $E(d_t) = 0$, where $d_t \equiv g(u_{1,t}) - g(u_{2,t})$ is the loss differential with a given loss function $g(\cdot)$. The DM statistic is computed as:

$$DM = \frac{\bar{d}}{\sqrt{2\pi \hat{f}_d(0)/T}}, \tag{31}$$

where

$$\bar{d} = \frac{1}{T}\sum_{t=1}^{T} \big(g(u_{1,t}) - g(u_{2,t})\big), \tag{32}$$

and $\hat{f}_d(0)$ is a consistent estimate of the spectral density of the loss differential at frequency 0.

$$WS = \sum_{t=1}^{T} I^+(d_t) \cdot \text{rank}(|d_t|), \tag{33}$$

where

$$I^+(d_t) = \begin{cases} 1, & d_t > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{34}$$

While loss functions provide a ranking of models based on average forecast accuracy, they do not indicate whether observed performance differences are statistically significant. To address this limitation, the Diebold–Mariano (DM) test. These tesst formally assess whether the difference in forecasting performance between two competing models is statistically distinguishable, rather than driven by sampling variability. Consequently, the DM test support model selection by verifying whether a model that performs better under the chosen loss function.

In the context of volatility forecasting, this additional validation is particularly important given the noisy nature of realized volatility measures.

5.4 Experimental Design

The experimental procedure was structured to systematically evaluate the forecasting performance of individual and hybrid volatility models. First, single-model benchmarks were estimated, including traditional GARCH-type models—namely GARCH, EGARCH, and EWMA—as well as standalone neural network models, including recurrent neural networks (RNN) and Long Short-Term Memory (LSTM) networks.

Next, hybrid models were constructed by integrating GARCH-type volatility forecasts with neural network architectures in order to assess whether combining econometric and deep learning approaches improves predictive performance. In particular, single hybrid models combined one GARCH-type model with an LSTM network. Building on this framework, double hybrid models were developed by combining two GARCH-type models with an LSTM. Finally, a fully integrated hybrid model was constructed by incorporating all three GARCH-type models—GARCH, EGARCH, and EWMA—into a single LSTM architecture, referred to as the GEW-GARCH model.

All GARCH-type models were estimated using the in-sample portion of the wheat futures data set, while forecast performance was evaluated using an out-of-sample testing period. This experimental design enables a comprehensive comparison of traditional econometric models, standalone deep learning models, and hybrid approaches to forecast future volatility.

Table 3: Estimated parameters of GARCH-type models

Model	Constant	ε_{t-1}^2	σ_{t-1}^2	Asymmetry
GARCH	0.032172	0.040428	0.949732	–
EGARCH	0.015577	0.090364	0.988610	0.021733
EWMA	0.03	0.97	–	–

The experimental design compares traditional econometric models, deep learning models, and hybrid approaches to assess the benefits of integrating GARCH-type volatility forecasts with LSTM networks.

6 Model Training and Hyperparameters

6.1 GARCH(1,1) Estimation

The GARCH(1,1) model is estimated by maximizing the log-likelihood function with respect to the parameter vector $(\mu, \omega, \alpha, \beta)$. Model estimation is carried out under the assumption of conditionally Gaussian innovations.

The optimization is performed using numerical maximum likelihood methods implemented in standard econometric software packages. The convergence criteria and optimization settings are summarized as follows:

- Optimizer: BFGS
- Maximum iterations: 10,000
- Convergence tolerance: 10^{-6}

Parameter estimates are retained only if the stationarity and positivity constraints of the GARCH model are satisfied.

6.2 RNN Training Setup

For the recurrent neural network (RNN)-based volatility models, including LSTM architectures, a consistent baseline training configuration is employed. Input sequences consist of rolling windows of past observations, and the models are trained to forecast future realized volatility.

The baseline configuration is summarized as follows:

- Architecture: Two stacked LSTM layers with 64 hidden units each
- Lookback window length: $L = 60$ trading days
- Output layer: Fully connected layer with softplus activation to ensure positive volatility forecasts
- Loss function: Mean squared error between predicted volatility $\hat{v}_{t+1}$ and realized volatility RV_{t+1}
- Optimizer: Adam with learning rate $\eta = 10^{-3}$
- Batch size: 64
- Regularization: Dropout with rate 0.2

Early stopping based on validation loss is employed to mitigate overfitting.

6.3 Hyperparameter Tuning

This study adopts a principled and parsimonious approach to hyperparameter selection, motivated by empirical robustness, interpretability, and comparability across model classes. Rather than performing extensive hyperparameter tuning, which risks overfitting in time-series settings, hyperparameters were chosen based on established theoretical guidance and prior empirical evidence in volatility forecasting.

Classical Volatility Models. For the GARCH family, a GARCH$(1, 1)$ specification was employed. This choice is well-supported in the literature, as the $(1, 1)$ model is known to capture the majority of volatility clustering observed in financial return series while remaining stable and interpretable. Higher-order specifications were deliberately avoided to prevent parameter redundancy and numerical instability. For EGARCH, an EGARCH$(1, 1)$ model with one leverage term ($o = 1$) was used to allow asymmetric responses to positive and negative shocks, consistent with the leverage effect documented in commodity markets. The EWMA model was implemented with a decay factor $\lambda = 0.97$, a standard value widely used in industry and academic applications, reflecting a balance between responsiveness and smoothing.

Neural Network Architectures. Neural models were intentionally kept shallow to reflect the limited effective sample size inherent in financial time series. The univariate MLP, SimpleRNN,

and LSTM architectures each employ a single hidden layer with a modest number of units, ensuring sufficient representational capacity while mitigating overfitting. Recurrent architectures were preferred over deep feedforward networks due to their ability to capture temporal dependence in volatility dynamics.

Lookback Window and Forecast Horizon. For LSTMs, the lookback length was chosen to be sufficiently long to capture short- and medium-term volatility persistence that comes to around $H = 60$ trading days. This around 25% of total trading days, having sufficient time to capture volatility clusters without introducing excessive noise from distant observations. The forecast horizon was fixed at $H = 22$ trading days, corresponding to the number of days within a month , and also the expiry time for near options. The volatilty predicted in this range will be of value to traders.

Training Procedure and Regularization. All neural networks were trained using the Adam optimizer with default learning rates, which provides adaptive step sizes suitable for noisy gradient environments. Early stopping based on validation loss was employed to prevent overfitting and ensure stable convergence. A fixed batch size was used across experiments to preserve coherency across training.

Evaluation Metrics. Model performance was assessed using both absolute and relative error measures, including RMSE, MAE, HMAE, and HMSE, calculated on the volatility realized in the daily-equivalent. Relative error metrics were emphasized to ensure scale invariance and robustness to volatility level shifts. Statistical significance of performance differences was evaluated using Diebold–Mariano tests based on the HMSE loss, to validate and provide a accurate comparision between performance of various models.

7 Results

Out-of-sample results on the test set are summarized in Table 4

Table 4: Out-of-Sample Forecast Performance Comparison

Model	**RMSE**	**MAE**	**HMAE**	**HMSE**
Hybrid LSTM Models				
GE LSTM (GARCH+EGARCH)	0.5584 ± 0.0059	0.3618 ± 0.0034	0.1929 ± 0.0024	0.0589 ± 0.0021
WE LSTM (EWMA+EGARCH)	0.5591 ± 0.0050	0.3634 ± 0.0035	0.1948 ± 0.0018	0.0597 ± 0.0016
GWE LSTM (GARCH+EGARCH+EWMA)	0.5602 ± 0.0071	0.3637 ± 0.0020	0.1949 ± 0.0033	0.0604 ± 0.0036
Hybrid LSTM + EGARCH σ^2	0.5626 ± 0.0080	0.3661 ± 0.0056	0.1949 ± 0.0044	0.0603 ± 0.0037
GW LSTM (GARCH+EWMA)	0.5628 ± 0.0049	0.3674 ± 0.0027	0.1962 ± 0.0034	0.0601 ± 0.0026
Hybrid LSTM + GARCH σ^2	0.5639 ± 0.0093	0.3670 ± 0.0074	0.1951 ± 0.0035	0.0598 ± 0.0025
Deep Learning Baselines				
LSTM (uni)	0.5659 ± 0.0107	0.3712 ± 0.0063	0.1968 ± 0.0054	0.0607 ± 0.0034
SimpleRNN (uni)	0.5934 ± 0.0137	0.4036 ± 0.0089	0.2148 ± 0.0075	0.0725 ± 0.0059
MLP (uni)	0.6196 ± 0.0108	0.3958 ± 0.0051	0.2008 ± 0.0016	0.0635 ± 0.0011
Econometric Baselines				
GARCH(1,1)	0.5917 ± 0.0000	0.3946 ± 0.0000	0.2283 ± 0.0000	0.0901 ± 0.0000
EWMA	0.6194 ± 0.0000	0.4140 ± 0.0000	0.2350 ± 0.0000	0.0936 ± 0.0000
EGARCH(1,1,o=1)	1.2014 ± 0.0000	1.0843 ± 0.0000	0.7102 ± 0.0000	0.6820 ± 0.0000

Hybrid LSTM models consistently outperform both classical and univariate neural benchmarks across all error metrics. The best-performing specification is the GE LSTM (GARCH+EGARCH), achieving the lowest RMSE of 0.558 ± 0.006, MAE of 0.362 ± 0.003, and HMAE of 0.193 ± 0.002,

representing a clear improvement over the univariate LSTM (RMSE 0.566, MAE 0.371). Hybrid LSTMs using EWMA and triple-feature combinations yield comparable but slightly inferior performance, indicating diminishing returns from adding highly correlated volatility signals. Hybrid RNN models improve upon univariate RNNs but remain consistently worse than LSTM-based hybrids. Classical models perform significantly worse, with GARCH(1,1) attaining an RMSE of 0.592 and EWMA 0.619, while EGARCH exhibits the poorest performance (RMSE 1.20, HMAE 0.71). Overall, results demonstrate that hybrid LSTM architectures combining one or two econometric volatility filters are seen to be most reliable for predicting short term volalitly trends.

7.1 Plots

For the sake of simplicity, we show 3 plots, one of the best model (RMSE: 0.5584), one with average case (RMSE: 0.5659) and the worst performing(RMSE: 1.201) as shown below. Figure 3. To check for the rest of the plots, they will be shown in the additional plot section at the end of the paper. Section A

8 Model Comparison and Robustness Checks

8.1 Model Comparison

Table 4 reports the out-of-sample performance of all models across four loss functions: RMSE, MAE, HMAE, and HMSE. These metrics jointly capture both squared-error sensitivity and robustness to large volatility forecast errors, which is particularly relevant in financial time series characterized by heavy tails.

Across all evaluation metrics, the *hybrid LSTM-based models* consistently outperform both the pure deep learning baselines and the traditional econometric models. In particular, hybrid specifications that incorporate conditional variance estimates from GARCH-family models as additional inputs achieve the lowest error values overall. Among these, the GE LSTM (GARCH+EGARCH) and WE LSTM (EWMA+EGARCH) variants exhibit the strongest and most stable performance across metrics.

In contrast, univariate deep learning baselines (LSTM, SimpleRNN, and MLP) perform noticeably worse, indicating that raw return histories alone are insufficient to capture the persistence and clustering effects present in volatility dynamics. Similarly, classical econometric models such as GARCH(1,1), EWMA, and EGARCH(1,1,o=1) underperform relative to hybrid models, highlighting the limitations of purely parametric volatility specifications when faced with nonlinear temporal dependencies. EGARCH exhibits the weakest forecasting performance in this study. While EGARCH is designed to capture asymmetric volatility responses, such leverage effects appear weak or unstable in the commodity futures data considered. Moreover, EGARCH operates in log-variance space, making multi-step forecasts highly sensitive to parameter estimation error and distributional assumptions.

Overall, these results suggest that explicitly combining econometric volatility structure with recurrent neural networks yields a superior bias–variance trade-off compared to either approach in isolation.

8.2 Robustness Checks

To assess robustness, each model was trained and evaluated over multiple independent runs using different random seeds, and the reported results correspond to the mean and standard deviation

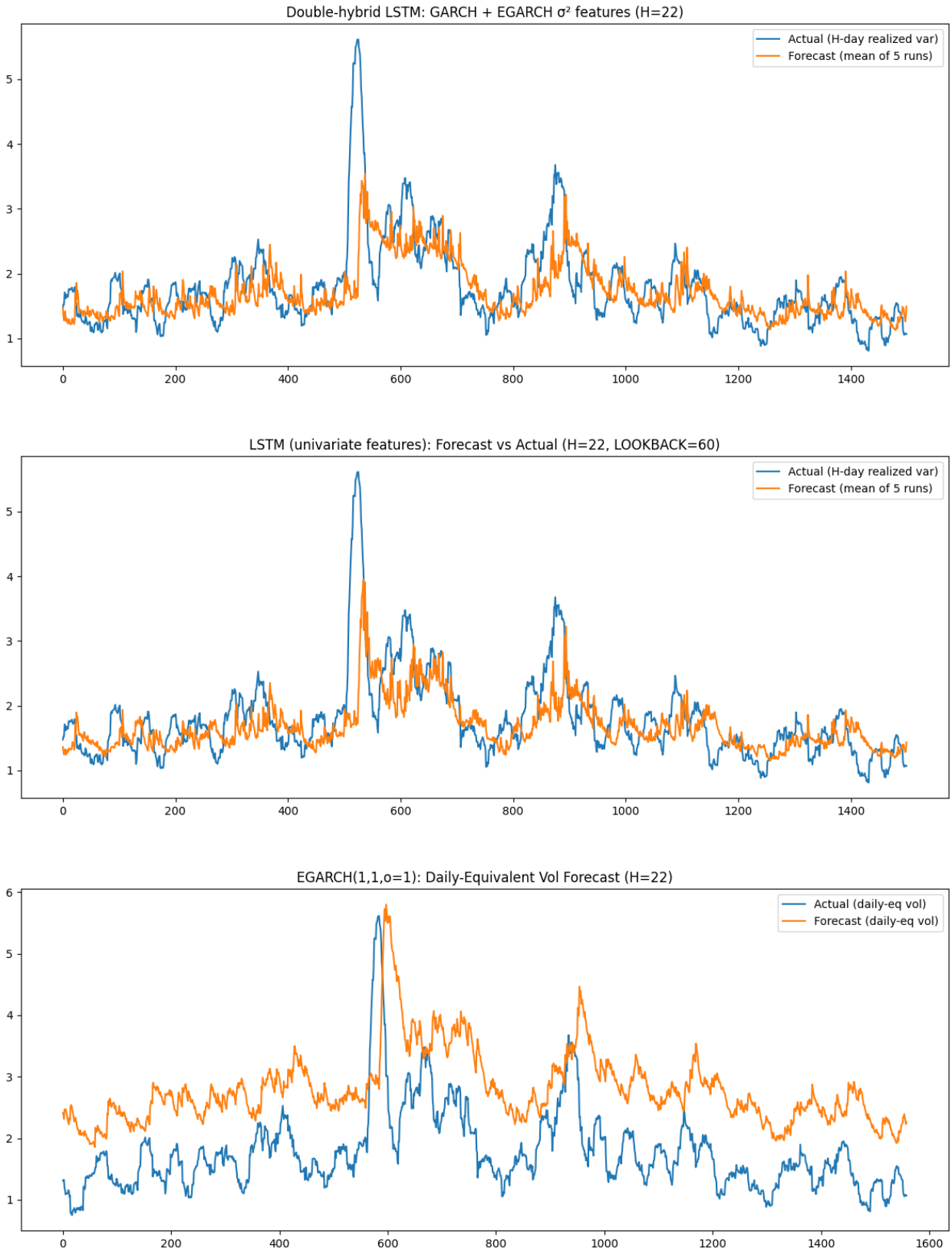


Figure 3: Performance comparison across best, intermediate, and worst-performing models

of the performance metrics. A key observation is that the hybrid LSTM models exhibit *very low standard deviations* across all error measures, typically an order of magnitude smaller than the corresponding mean values.

The low dispersion in performance indicates that the hybrid models are not overly sensitive to random initialization or stochastic optimization effects. This stability suggests that the gains observed are structural rather than incidental, arising from the inclusion of economically meaningful volatility features rather than favorable random draws.

By contrast, univariate deep learning models display higher variability across runs, reflecting greater sensitivity to initialization and potential overfitting. Econometric models show zero standard deviation by construction, as they are deterministic given the data; however, their inferior average performance underscores that stability alone is insufficient without predictive accuracy.

Taken together, the consistently low variance and superior mean performance of the hybrid models provide strong evidence of robustness. These findings support the conclusion that hybrid LSTM–GARCH architectures offer a reliable and stable framework for volatility forecasting, capable of delivering both improved accuracy and repeatable results across multiple training runs.

9 Risks & Limitations

While the proposed models demonstrate strong forecasting performance, several risks should be acknowledged. Classical volatility models such as GARCH and EGARCH rely on strong distributional and stationarity assumptions, which may be violated during structural breaks or extreme market events. Neural network models, particularly LSTMs, are susceptible to overfitting and may learn spurious temporal patterns when trained on limited financial time-series data. Hybrid models inherit risks from both components, including error propagation from econometric volatility filters and increased sensitivity to feature collinearity. Multi-step volatility forecasts further rely on expectation-based approximations, which can accumulate error over longer horizons. Finally, model performance may degrade under regime shifts, changes in market microstructure, or rare tail events not well represented in the training sample.

10 Challenges Faced

- **Target Definition and Scale Consistency:** Defining an appropriate volatility target required careful consideration of the number of days required to predict. Ensuring consistent scaling across classical, neural, and hybrid models was non-trivial and necessitated multiple validation checks.

- **Data Preparation:** It took time to realize that commodity futures exhibit high volatility near expiry, hence getting adjusted data that could exhibit the volatility dynamics we intended to capture took a long time.

- **Model Comparison and Metric Selection:** Selecting evaluation metrics suitable for volatility forecasting, while avoiding misuse of variance-based losses such as QLIKE for volatility targets, required careful methodological adjustments and re-evaluation of model performance.

- **Training Stability and Overfitting Risk:** Neural network models exhibited sensitivity to hyperparameter choices and training procedures, necessitating early stopping, constrained architectures, and multiple runs to ensure stable and reproducible results.

11 Conclusion

The evaluation of different volatility forecasting models demonstrates that **hybrid LSTM architectures consistently outperform both classical models and univariate neural networks** across all error metrics. Among the hybrids, the **GE LSTM (GARCH + EGARCH)** achieves the best results, with an RMSE of 0.558, MAE of 0.362, and HMAE of 0.193, showing clear improvement over univariate LSTMs. While hybrid models incorporating EWMA or multiple correlated features offer comparable performance, they provide only marginal gains, highlighting diminishing returns from adding highly correlated signals. Hybrid RNNs improve upon univariate RNNs but remain inferior to LSTM-based hybrids. Classical econometric models lag significantly behind, with EGARCH performing the worst. **Overall, parsimonious hybrid LSTM models that integrate one or two econometric volatility filters provide the most accurate and stable forecasts**, making them the preferred choice for volatility prediction.

A Additional Figures

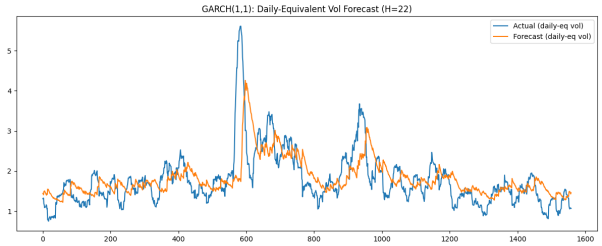


Figure 4: GARCH Volatility Plot

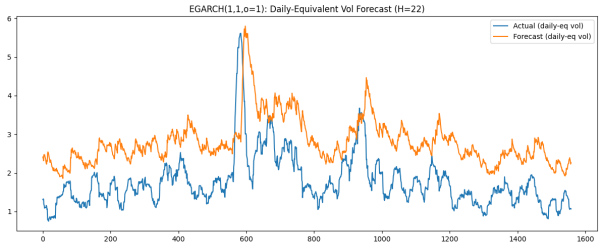


Figure 5: EGARCH Volatility Plot

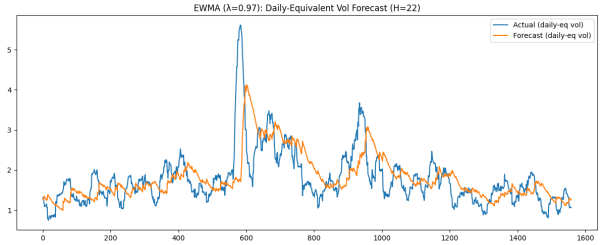


Figure 6: EWMA Volatility Plot

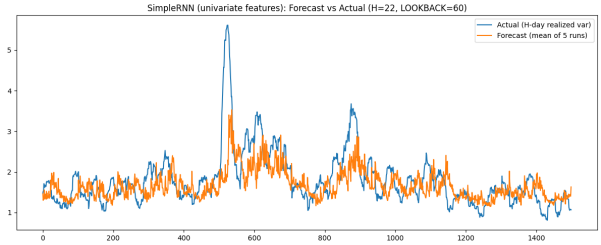


Figure 7: Simple RNN Forecast

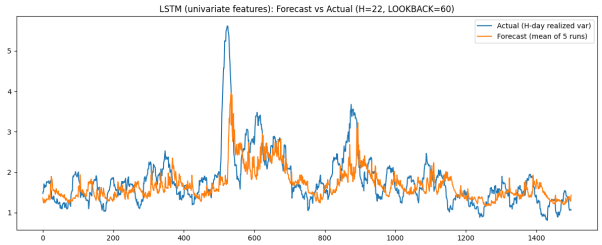


Figure 8: LSTM Forecast

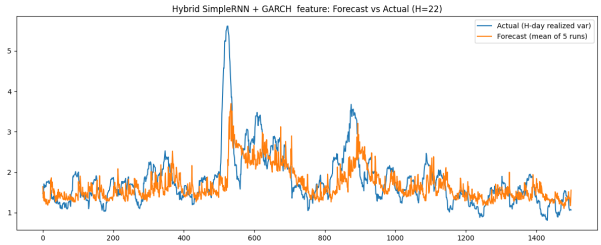


Figure 9: Hybrid RNN-GARCH

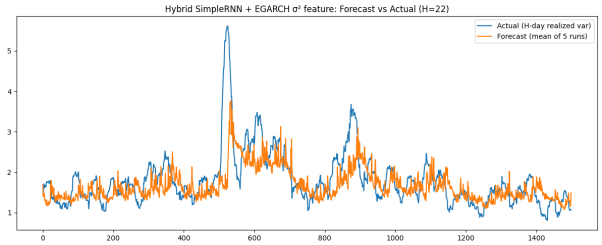


Figure 10: Hybrid RNN-EGARCH

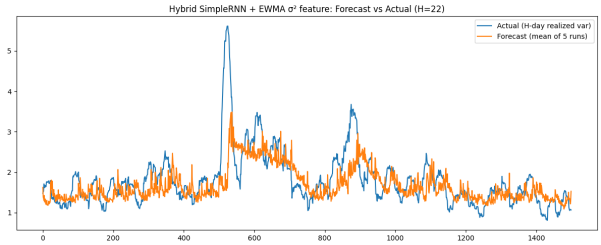


Figure 11: Hybrid RNN-EWMA

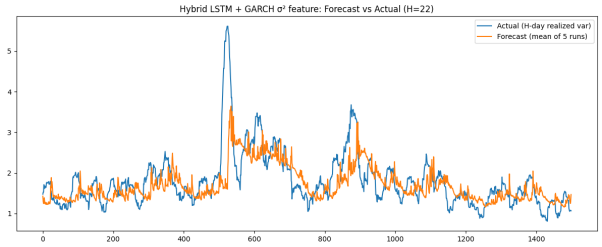


Figure 12: Hybrid LSTM-GARCH

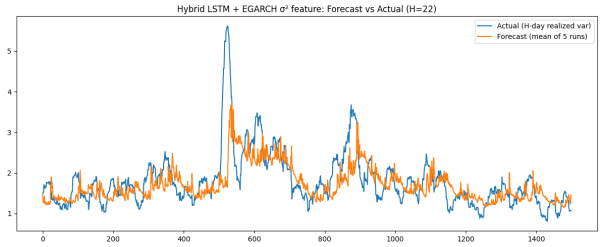


Figure 13: Hybrid LSTM-EGARCH

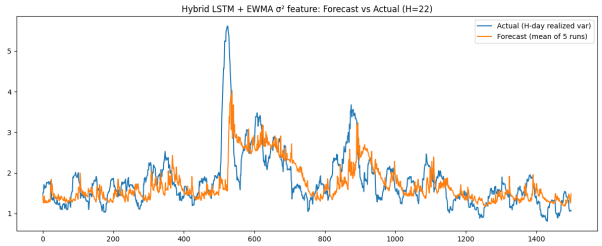


Figure 14: Hybrid LSTM-EWMA

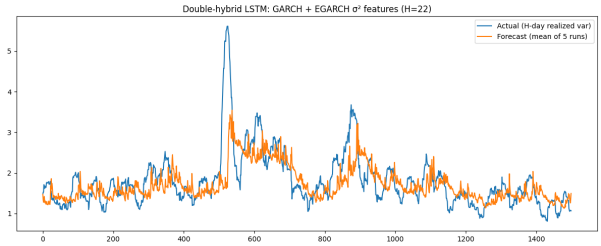


Figure 15: LSTM-GARCH-EGARCH

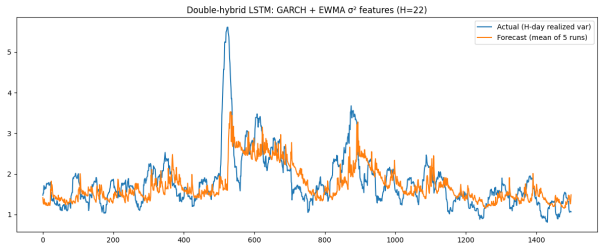


Figure 16: LSTM-GARCH-EWMA

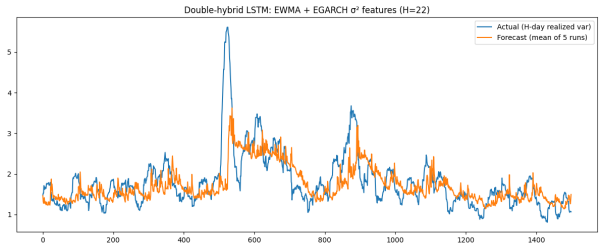


Figure 17: LSTM-EWMA-EGARCH

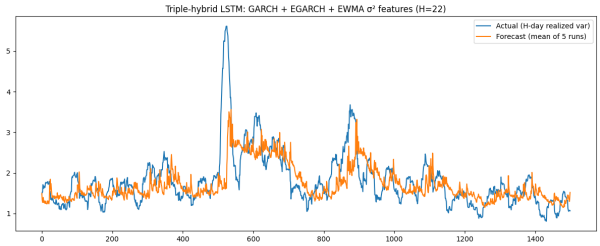


Figure 18: Triple Hybrid Model

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